School of Accounting & Finance

FORM FOR THE SUBMISSION OF ASSESSED GROUP COURSEWORK

THIS FORM MUST BE COMPLETED AND ATTACHED TO **ALL** ASSESSED GROUP WORK.

Please include all group members' student numbers below.

Student Number	2387326
Student Number	2496574
Student Number	2478440
Student Number	2383228
Student Number	2496713

We agree that equal and fair contributions were made to the final report by all members of our group.

* Please delete as appropriate. In the event that all members of the group do not agree that they have worked together effectively, shared the workload equitably and are therefore happy to receive the same individual mark for the project, they will be required to provide written evidence as to why this occurred.

Please list brief details of how tasks were divided up among the group members in the box below (please refer to chapters of final report etc.):

Abstract- 2383228
Introduction-2478440
Literature Background-2478440
Data and Data collection from Eikon- 2387326
Methodology-2496713
Result- 2387326, 2496574
Conclusion-2383228, 2496574
Latex Reporting and Format-2387326
Latex Tables and figures-2387326

By submitting this work online using my unique log-in and password, I confirm that I am submitting the work on behalf of my group.

We understand that all marks are provisional until ratified by the Faculty Examination Board.

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ASSESSMENT FEEDBACK FORM
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Word Count- 3577

Unit Title: Empirical Finance

Unit Code: EFIMM0046

Student Number: 2387326, 2496574, 2478440, 2383228, 2496713

MARK*

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Please start writing your coursework on the next page.

Do Active Funds Perform Better Than Passive Funds ?

Abstract

Our paper attempts to evaluate the performances of Actively Managed Mutual Funds and Passive (Index) Mutual Funds for funds notified for sale only in the US and the UK over the period of April 2019 to April 2024 empirically. Here, we try to assess the objective of "Do actively managed mutual funds been able to generate statistically superior returns as compared to passive funds ?". Key performance metrics like Jensen's Alpha, Beta, Sharpe Ratio, Treynor's Ratio, and fund returns for 502 Active funds and 37 Passive funds have been analysed. The results indicate that active funds generally have higher returns and volatility but underperform their benchmarks more significantly than passive funds, as evidenced by a more negative average Jensen's Alpha. Despite slight advantages in Sharpe and Treynor Ratios suggesting better risk-adjusted returns for active funds, these differences are not statistically significant, challenging the efficacy of higher fees associated with active management. The study robustly rejects the null hypothesis for returns, with a p-value of 0.00733 indicating statistically significant higher returns for active funds. However, the analysis faces limitations such as a small sample size for passive funds and potential biases arising from inconsistent data due to birth of new funds and death of existing active funds for the observed sample period.

1 Introduction

The effectiveness of active versus passive fund management strategies is a topic of constant debate in the field of investment management. Passive funds attempt to replicate the performance of a benchmark index with limited trading and lower costs. Active mutual funds, on the other hand, are operated by professionals who proactively select stocks and seek to outperform the market.

The discussion has major practical consequences on managing portfolios, primarily with regard to risk management and possible yields. As more investors opt for passive investments, that offer lower fees and often yield superior or identical long-term outcomes, active funds are under greater pressure to justify the reason they charge higher costs and management fees.

A broad range of findings have been obtained from previous study in this area. Some studies indicate that active management may perform more efficiently than passive strategies in some market scenarios, consisting significant amounts of volatility or market correction itself. This is because active managers are possibly more equipped to deal with fluctuations in the market (Jones and Wermers, 2012). On the contrary, another study suggests markets continue to grow more efficient and it's challenging for active managers to consistently surpass their benchmarks (Fama and French, 2010). Even after a lot of research, there still exists no straightforward response because the approaches used, the time intervals considered at, and the market segments evaluated varied. Additionally, survivorship bias and fund selection may influence outcomes significantly.

A large set of data from both active and passive equity mutual funds is utilized in this study in order to make the theory explicit. The aim of this study is to look at performance over five years by analyzing 502 active funds and 37 passive funds obtained from Eikon Database. Statistical methods like Jensen's alpha, Sharpe ratios, and CAPM betas will be implemented to construct a complete analysis of risk-adjusted returns.

The main objective of the research is to ensure that investors and financial experts recognize if active funds' performance strategies are certainly superior to passive ones, taking into account various market circumstances and time periods. With extensive study, our research examines the main question: Do active funds perform better than passive funds?

2 Literature Background

Both active and passive fund management constitute significant topics of argument in finance, and they are thoroughly examined in a range of market circumstances. Recent studies have introduced additional details to this argument. One important study has been conducted by Ľuboš Pástor and M. Blair Vorsatz, who evaluated U.S. stock mutual funds' performance during the COVID-19 crisis. Atanu Saha and Alex Rinaudo's study, on the other hand, compared large actively managed funds with passively managed funds in a variety of market segments. Their results showed that active funds might actually do better than inactive ones, especially when it comes to average returns, risk-adjusted returns, and managing downside risks. They used both statistical analysis and Monte-Carlo simulation to scrutinize the fund log returns and measure success. This study demonstrates that active fund management may work better or worse based on the nature of market and the fund managers' particular competencies.

The study by Klemens Kemnitzer on emerging markets adds to the debate about active vs. passive fund management. Kemnitzer says that active funds might have an edge in markets that aren't very efficient because it may utilize arbitrage to gain advantage. On the other hand, the paper stresses how crucial it is to take into account tax implications and other fund characteristics when gauging success, since these can have a large impact on net investor returns.

Alan D. Crane and Kevin Crotty disagree that active management is frequently an optimal idea. According to their study, some active funds might exhibit better performance when considering gross alphas, but this advantage isn't as notable when you consider passive fund performance which is spread out and residual risk is factored into account. They concluded that these adjustments make the performance variances among active and passive funds nearly eliminate. This implies that passive funds might offer higher quality yields per unit of residual risk.

From 1995 to 2008, Dale A. Prondzinski completed a study illustrating the effect of market cycles on the performance of funds. Based on the Sharpe composite portfolio performance measure, Prondzinski asserts that passive management may perform more effectively when the market improves, however the opposite could be true when the market collapses. It's clear from this study that fund achievements experiences cycles, and it appears like market timing could affect the decision between active and passive management. In a landmark study published in 2010, Fama and French discovered that most active fund managers fail to perform more effectively over time than passive fund managers, regardless of fees are considered. They studied the returns of various mutual funds in relation to norms that adopted differences in risk and style. They determined that active funds frequently incur higher transaction costs and administration fees that balance out any additional returns that are generated by active fund management.

"The Arithmetic of Active Management," which Sharpe wrote in 1991, is a quantitative work that advances the idea that passive fund typically performs better than active fund owing to differences in costs. He stated that prior to costs, the average dollar that is actively tracked should make the identical return as the average dollar that is managed passively. When costs are included, though, the return on the average actively managed currency will be less than the return on the passive currency. In 2003, Malkiel backed up the Efficient Market Hypothesis by claiming that active managers can't consistently perform better than the overall market because the market is capable of taking in new information. His study claims the prices of stocks already display all the information that is readily accessible, active fund is not bound to deliver superior returns that prevail.

Whereas there is proof that active fund managers might have an edge in some markets or sectors that are not operating well. Sorensen, Miller, and Samak (2000) observed into this notion and concluded that active fund management could potentially perform better than passive fund techniques in markets where information is sparse and equities have a higher probability to be mispriced.

To give an extensive overview of the way the performance of active and passive funds changes over time, this literature review incorporates these fundamental research findings in addition to more recent statistical approaches. It reinforces up the belief that passive funds generally have a beneficial cost-to-performance ratio than active funds, despite the fact that active management may be more effective in specific circumstances. This is particularly applicable in well-established, profitable markets over the long run.

3 Data

Data is collected from Refinitiv Thomson Eikon Fundscreener. The asset universe used is mutual funds that is notified for sale only in the US and UK. The funds chosen are equity funds that invest in equities only in emerging markets as well as overall equities market globally. Funds are currently alive and investors can currently buy its shares, which are later used in purchasing and selling assets, managing investor funds.

The frequency of observations is Monthly. Total Return Index for last 5 years i.e) from April 2019 to April 2024 is studied using Datastream. It is used because its adjusted for dividends and stock splits. Return index for last 5 years is used since its a holistic measure of performance reflecting both change in Net Asset Value (NAV) and reinvestment of distributions.

Filtering out, we have 502 actively managed mutual funds. Duplicates avoided and only independent observations included. In order for the sample chosen to be representative of entire population being studied (Selection Bias) and to avoid focusing only on surviving funds, we take into account both the alive and dead funds. This eliminates the possibility of selection bias as well survivorship bias.

The next dataset includes 37 Passive (Index) funds. The frequency of observations is Monthly. The total Return Index for the last 5 years from April 2019 to April 2024 is obtained. The funds aim to replicate the performance of specific market indexes by holding the same assets in the same proportions. Passive funds have lower fees compared to actively managed mutual funds.

Winsorization: Both the Total Return Index of Active and Passive funds are capped at the 5th and 95th percentiles which helps in significantly reducing the influence of outliers which can affect statistical analyses and lead to misleading bloated conclusions. 3-month US Treasury Bill Rate - 5.25 percent.

The benchmark index against which fund performance is measured is MSCI (Morgan Stanley Capital International) EMERGING MARKETS NR USD.

MSCI EMERGING MARKETS USD	Market Index Returns Average Annual (%)
Mean	12.87
Standard Deviation	19.05

Table 1: Market Index Return Average Annualized

4 Methodology

To evaluate "Do active funds perform better than passive funds?", we use the performance metrics (Sharpe ratio, Treynor's ratio and Jensen's Alpha) which help in assessing active and passive fund's performance relative to their risk and benchmark, and Statistical tests (Welch's t-test) make it clear to examine the variations in returns active and passive funds while simultaneously making certain that any differences viewed are statistically significant and not simply spurious variation.

Sharpe Ratio:. The Sharpe Ratio measures the performance of an investment compared to a risk-free asset, after adjusting for its risk. It is calculated as the difference between the returns of the investment and the risk-free rate, divided by the standard deviation of the investment's returns. A higher Sharpe Ratio indicates better risk-adjusted returns, making it a crucial tool for comparing the effectiveness of different investments.

$$S = (\frac{(Rp - Rf)}{\sigma p})$$

Treynor Ratio:. Treynor's Ratio is a way to gauge the efficiency of a financial fund by comparing the return on investment to its market risk. When you compare the funds' return over the risk-free rate to its beta, it determines the extra return made for every unit of risk undertaken. A higher Treynor's Ratio signals enhanced risk-adjusted performance, which implies that the fund has higher returns per unit of market risk in comparison to funds with a lower Treynor's ratio. Therefore, Treynor's Ratio can be extremely beneficial for examining fund performance, even though they are subjected to various quantities of market risk.

$$Treynor'sRatio = \frac{(Rp - Rf)}{\beta}$$

Jensen Alpha:. Jensen's Alpha is a financial metric that monitors this difference which effectively examines if a fund can produce higher profits than the benchmark considering the risk of the fund, which can be determined by its beta. If Jensen's Alpha is positive, it suggests that the fund performed superior to its benchmark, and that is an indication of competent management. If alpha is negative, it denotes that the fund failed to perform as well as it should have. Considering the potential hazards, this metric is crucial for determining fund performance at beating market projections.

$$Jensen'sAlpha = (Rp - (Rf + \beta(Rm - Rf)))$$

Welch's t-test:. Welch's t-test is a statistical method used to determine if there is a significant difference between the mean values of the two categories when the groups have different variances and possibly even different sample sizes. This renders it an excellent tool for contrasting things like outcomes between active and passive funds. Welch's t-test varies the degrees of freedom depending on the sample variances and sizes, providing a more realistic representation of significance when these assumptions aren't met. This differs from the standard t-test, which assumes that the variances between the groups are equal. For financial data, this works particularly well because active and passive funds can possess extremely distinct levels of investment volatility and their sample sizes can be quite distinct.

5 Results and Discussion

TABLE 01: Descriptive Statistics: Performance Measures of Active Funds

Fund Performance Measures	Jensen's Alpha	CAPM Beta	Sharpe Ratio	Treynor Ratio	Fund Returns (Total Return Index)
Mean	-0.190918	0.978444	0.062824	0.340387	207.574015
Median	-0.18033	0.987629	0.071151	0.392665	202.792320
Standard Deviation	0.231861	0.110576	0.071190	0.385951	38.261836
Kurtosis	1.163560	3.751663	-0.729298	-0.863997	-0.491298
Skewness	-0.4381385	-0.253945	-0.167785	-0.277454	0.017096
Min	-1.092029	0.495629	-0.125291	-0.548232	165.164113
25%	-0.314358	0.927905	0.000790	0.004490	183.667355
50%	-0.180338	0.987629	0.071152	0.392666	202.792320
75%	-0.048728	1.040173	0.119121	0.650057	222.823320
Max	0.558213	1.527715	0.278068	1.266085	292.198905

This table provides summary statistics for various fund performance measures to assess the performance of actively managed mutual funds. The performance measures include Jensen's Alpha, CAPM Beta, Sharpe Ratio, Treynor Ratio and Fund Returns. The Fund Returns are denoted by Total Return Index obtained from Refinitiv Eikon using FundScreener.

TABLE 02: Descriptive Statistics: Performance Measures of Passive Funds

Fund Performance Measures	Jensen's Alpha	CAPM Beta	Sharpe Ratio	Treynor Ratio	Fund Returns (Total Return Index)
Mean	-0.057285	0.958650	0.045400	0.23856	152.052977
Median	-0.061844	0.997807	-0.00188	-0.01039	151.494459
Standard Deviation	0.086361	0.110605	0.08542	0.45513	17.574109
Kurtosis	6.784903	4.772463	-1.36021	-1.29203	-0.219292
Skewness	2.2491833	-2.563723	0.545955	0.56592	-0.118852
Min	-0.183239	0.637687	-0.07379	-0.38324	122.780486
25%	-0.087905	0.988982	-0.00911	-0.05068	139.558311
50%	-0.061844	0.997807	-0.00188	-0.01039	151.494459
75%	-0.045996	1.001936	0.14640	0.765989	166.432703
Max	0.254254	1.023009	1.19829	1.09316	180.638824

This table provides summary statistics for various fund performance measures to assess the performance of Passive Funds. The performance measures include Jensen's Alpha, CAPM Beta, Sharpe Ratio, Treynor Ratio and Fund Returns. The Fund Returns are denoted by Total Return Index obtained from Refinitiv Eikon using FundScreener.

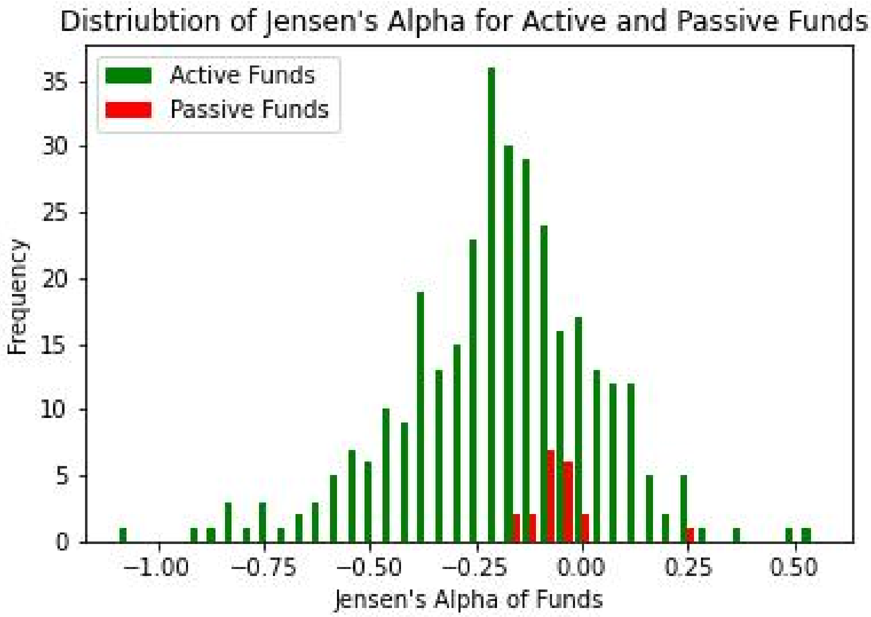

Fig. 1: Histogram of Calculated Jensen's Alpha of Active and Passive Funds and its comparison. It can be noted from the above histogram that the distribution is bell-shaped .

JENSEN'S ALPHA

Two-sample t-test assuming unequal variances (Welch T-Test)	Jensen's Alpha: Actively Managed Mutual Funds	Jensen's Alpha: Passive Index Funds
Mean	-0.190918	-0.057285
Standard Deviation	0.231861	0.086361
P(T<=t) Two-Tail (p-value)	p<0.0001	

Table 4: Jensen's Alpha Test Results For Difference Between Actively Managed Mutual Funds And Passive Funds

According to the 5-year data (April 2019 – April 2024) of US and UK active funds (502) with a Jensen's Alpha range of -1.09 to 0.56 and passive funds (37) with a Jensen's Alpha range of -0.18 to 0.25, the mean Alpha of the passive is slightly negative (-0.057), which indicates that on average the passive funds did not outperform their benchmarks. The mean Alpha of the active funds is significantly more negative (-0.1909), suggesting active funds underperformed their benchmarks more compared to passive funds. Active funds have a more negative average Alpha than passive funds, indicating worse performance relative to their benchmarks.

Null Hypothesis (H0): There is no significant difference between Jensen's Alpha of actively managed mutual funds and passive funds.

Alternative Hypothesis (H1): There is a significant difference between Jensen's alpha of actively managed mutual funds and passive funds.

The p-value ($p < 0.0001$) is significantly less than threshold alpha of 0.05. This indicates strong evidence against the null hypothesis. Thus, we reject the null hypothesis and conclude there is statistically significant difference between the Jensen's Alpha measures of actively managed mutual funds and passively managed funds. Active funds (-0.190) underperform more than passive funds (-0.057).

CAPM BETA

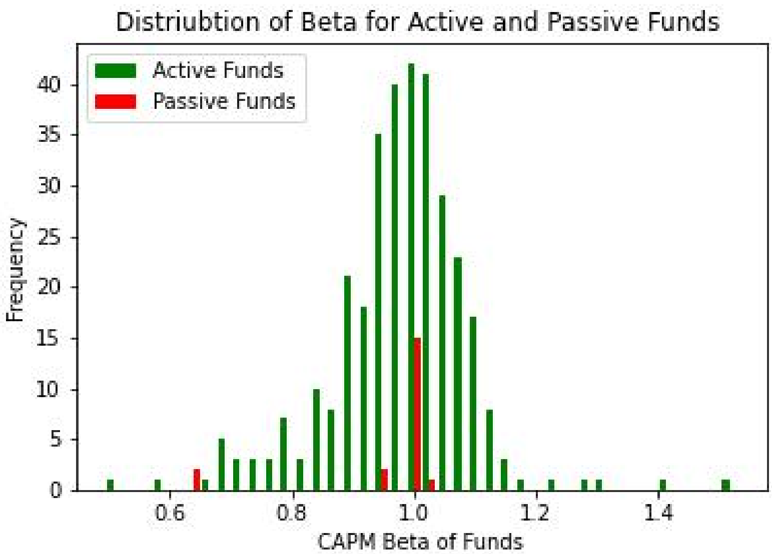

Fig. 2: Histogram of Calculated CAPM Beta of Active and Passive Funds and its comparison. It can be noted from the above histogram that the distribution is bell-shaped .

	CAPM Beta	
Two-sample t-test assuming unequal variances (Welch T-Test)	Actively Managed Mutual Funds	Passive Index Funds
Mean	0.9784	0.9586
Standard Deviation	0.1105	0.1106
P(T¡=t) Two-Tail (p-value)	p = 0.4456	

Table 5: CAPM Beta Test Results For Difference Between Actively Managed Mutual Funds And Passive Funds

According to the 5-year data (April 2019 – April 2024) of US and UK active funds (502) with a Beta range of 0.496 to 1.528 and passive funds (37) with a Beta range of 0.64 to 1.02, the average Beta of the passive funds is close to one (0.9587). The average Beta for active funds is close to one (0.9784) indicating a strong correlation with market movements. Both types of funds exhibit Beta values close to one which indicates that market movements strongly influence their returns. This similarity reflects that both the funds do not manage to differentiate themselves significantly in terms of market-related risk.

Null Hypothesis (H0): There is no significant difference between Beta of actively managed funds and passive funds.

Alternative hypothesis (H1): There is significant difference between Beta measures of actively managed funds and passive funds.

Morever the p-value is 0.4456 which is greater than alpha of 0.05. Hence we fail to reject the null-hypothesis. There doesn't seem sufficient evidence to conclude that the two types of funds have considerable variations in beta.

Sharpe Ratio

	Sharpe Ratio	
Two-sample t-test assuming unequal variances (Welch T-Test)	Actively Managed Mutual Funds	Passive Index Funds
Mean	0.0628	0.0454
Standard Deviation	0.0711	0.0854
P(T<=t) Two-Tail (p-value)	p = 0.3819	

Table 6: Sharpe Ratio Test Results For Difference Between Actively Managed Mutual Funds And Passive Funds

According to the 5-year data (April 2019 – April 2024) of US and UK active funds (502) with a Sharpe ratio range of -0.13 to 0.28 and passive funds (37) with a Sharpe ratio range of -0.07 to 0.198, the average Sharpe Ratio of the active funds (0.063) is slightly higher than the average Sharpe ratio of the passive funds (0.045), which indicates that active funds have slightly higher risk-adjusted returns than passive funds. The difference between the funds is not significant as well as it might not justify the higher fees associated with active fund managers.

Null hypothesis (H0): The Sharpe ratio is the same for both active and passive funds.

Alternative hypothesis (H1): The Sharpe ratio is drastically distinct between both active and passive funds.

The p-value of 0.3819 stipulates that the null hypothesis cannot be rejected since it is greater than alpha=0.05, which implies that there is no significant difference in the Sharpe ratio between the active and passive funds.

Treynor's Ratio

	Treynor's Ratio	
Two-sample t-test assuming unequal variances (Welch T-Test)	Actively Managed Mutual Funds	Passive Index Funds
Mean	0.3403	0.2385
Standard Deviation	0.3859	0.4551
P(T<=t) Two-Tail (p-value)	p = 0.3387	

Table 7: Treynor's Ratio Test Results For Difference Between Actively Managed Mutual Funds And Passive Funds

According to the 5-year data (April 2019 – April 2024) of US and UK active funds(502) with a Treynor's ratio range of -0.55 to 1.27 and passive funds(37) with a Treynor's ratio range of -0.38 to 1.09, the mean Treynor's Ratio of the active funds (0.3403) is higher than the mean Treynor's ratio of the passive funds(0.2386) which suggests that active funds have better returns per unit of market risk than passive funds which suggest that active funds offer higher compensation than the passive funds for the risk related to the market volatility.

Null Hypothesis (H0): There is no significant difference between Treynor's Ratios of actively managed mutual funds and passive funds.

Alternative Hypothesis (H1): There is a significant difference between Treynor's Ratios of actively managed mutual funds and passive funds.

Employing p-value of 0.3387 which is greater than alpha=0.05, we fail to reject the null hypothesis. Hence, there isn't persuasive significant evidence that the Treynor ratio varies within the two categories of funds.

Fund Returns (Total Return Index)

	Fund Returns as Total Return Index (RI)	
Two-sample t-test assuming unequal variances (Welch T-Test)	Actively Managed Mutual Funds	Passive Index Funds
Mean	207.5740	152.0529
Standard Deviation	38.2618	17.5741
P(T<=t) Two-Tail (p-value)	p = 0.0073	

Table 8: Fund Returns Test Results For Difference Between Actively Managed Mutual Funds And Passive Funds

According to the 5-year winsorized data (April 2019 – April 2024) of US and UK active funds (502 funds) with a mean total return index range of 165.26 to 292.32 and passive funds (37 funds) with a range of 124.23 to 181.55, the active funds have higher returns, more volatility, and risk compared to passive funds, whereas the passive funds provide more consistent and stable returns.

Null Hypothesis (H0): There is no significant difference in returns between active and passive funds

Alternative Hypothesis (H1): There is a significant difference in returns between active and passive funds.

The T-statistic figure of 2.70267 illustrates that the average returns of active funds exceed passive funds. Since the t-statistic is positive, this indicates that active funds generate higher average returns than passive funds.

Additionally, there is strong evidence against the null hypothesis considering the p-value of 0.00733 is less than 0.05. Hence, we reject the null hypothesis and the difference is statistically significant. Based on the above results, we observe active funds perform slightly better than passive funds in terms of fund returns and the difference between them is highly significant.

6 Conclusion

The average Jensen's Alpha for active funds was greater than that of passive funds, which suggests they performed negatively when compared with benchmarks. Similar Betas for both types of funds demonstrated that they were equally sensitive to fluctuations in the market, without any substantial variations in the degree of market risk. Even though Sharpe and Treynor Ratios were slightly larger for active funds, which may suggest better risk-adjusted returns and higher gains per unit of market risk, these differences were not statistically significant. Risk-adjusted metrics when factored into account, the evidence fails to convincingly demonstrate that active funds perform better than passive funds.

Active funds have high return index and the difference of returns between the funds is substantially significant. A slightly more favorable return on risk (though the difference is not significant) and Beta being more unpredictable for active funds and its underperformance comapared with passive funds (Jensen's Alpha), passive funds' performance seem to be better than active funds .As a result, active funds may not always be as successful as passive funds concerning overall performance, particularly when fees are considered. The active funds performed better than passive funds only on account of Total Return Indices. For Sharpe and Treynor's Ratio there was no significant difference in the performance of actively managed mutual and passive funds.

This means that selecting between active and passive funds should depend on the risk tolerance of the investor, their investment goals, and their preference for potential higher returns versus stability and lower fees.

Limitations.

Usage of Fund Return Index as a performance measure. When using Return Indices as a performance measure, some active funds possibly could have died and some new funds might have originated within the sample time period. This results in bloated mean return indices and leads to biased and inaccurate results.

Small Sample Size:. The number of passive funds (37) compared to active funds (502) is disproportionately small. This could skew the results and make it challenging to generalize findings to a broader population of funds.

Statistical Significance:. While p-values are provided, the significance levels used (0.05) may not be appropriate for all comparisons. Adjusting for multiple comparisons or employing stricter significance levels might alter the conclusions.

Assumption of Normality:. The analysis assumes that the data follow a normal distribution, which might not be true for financial data. Non-normality could affect the accuracy of statistical tests and the reliability of the conclusions.

Simplistic Hypotheses: . The null and alternative hypotheses are oversimplified and may not capture the complexity of fund performance. For instance, focusing solely on whether alpha or beta are different between active and passive funds overlooks other potential differences.

No Control for Risk Factors:. The analysis doesn't control for other risk factors that could influence fund performance, such as sector exposure, style biases, or management strategies. Ignoring these factors could lead to biased results.

Lack of Attribution Analysis:. The analysis doesn't break down the sources of fund performance. Understanding which factors contribute to returns would provide deeper insights into why active or passive funds perform differently.

Risk-Return Trade-off:. While active funds may have higher returns, they also exhibit more volatility and risk compared to passive funds. This trade-off is not fully explored, and risk-adjusted measures could provide a more balanced comparison. Addressing these limitations would improve the reliability and robustness of the analysis, providing more accurate insights into the performance of active and passive funds.

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